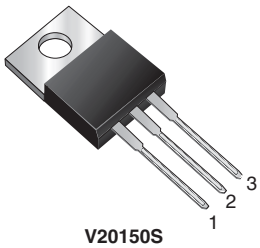
V20150S-E3, VF20150S-E3, VB20150S-E3, VI20150S-E3

High Voltage Trench MOS Barrier Schottky Rectifier

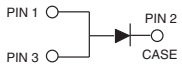
Ultra Low V_F = 0.55 V at I_F = 5 A

TMBS®

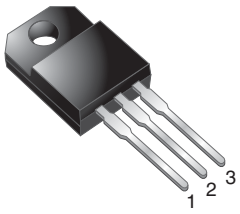
TO-220AB



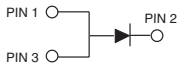
V20150S



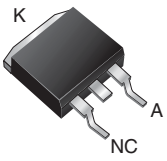
ITO-220AB



VF20150S



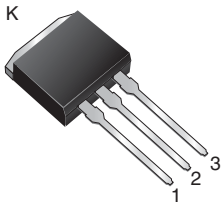
D²PAK (TO-263AB)



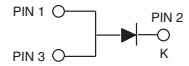
VB20150S



TO-262AA



VI20150S



DESIGN SUPPORT TOOLS

click logo to get started

3D Models Available

FEATURES

- Trench MOS Schottky technology
- Low forward voltage drop, low power losses
- High efficiency operation
- Meets MSL level 1, per J-STD-020, LF maximum peak of 245 °C (for TO-263AB package)
- Solder bath temperature 275 °C maximum, 10 s, per JESD 22-B106 (for TO-220AB, ITO-220AB, and TO-262AA package)
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

Pb e3 RoHS COMPLIANT

TYPICAL APPLICATIONS

For use in high frequency converters, switching power supplies, freewheeling diodes, OR-ing diode, DC/DC converters and reverse battery protection.

MECHANICAL DATA

Case: TO-220AB, ITO-220AB, D²PAK (TO-263AB), and TO-262AA

Molding compound meets UL 94 V-0 flammability rating Base P/N-E3 - RoHS-compliant, commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
E3 suffix meets JESD 201 class 1A whisker test

Polarity: as marked

Mounting Torque: 10 in-lbs max.

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	20 A
V_{RRM}	150 V
I_{FSM}	160 A
V_F at I_F = 20 A	0.75 V
T_J max.	150 °C
Package	TO-220AB, ITO-220AB, D²PAK (TO-263AB), TO-262AA
Circuit configuration	Single

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER	SYMBOL	V20150S	VF20150S	VB20150S	VI20150S	UNIT
Max. repetitive peak reverse voltage	V_{RRM}	150				V
Max. average forward rectified current (fig. 1)	$I_{F(AV)}$	20				A
Peak forward surge current 8.3 ms single halfsine-wave superimposed on rated load	I_{FSM}	160				A
Non-repetitive avalanche energy at T_J = 25 °C, L = 60 mH	E_{AS}	150				mJ
Peak repetitive reverse current at t_p = 2 µs, 1 kHz, T_J = 38 °C ± 2 °C	I_{RRM}	0.5				A
Voltage rate of change (rated V_R)	dV/dt	10 000				V/µs
Isolation voltage (ITO-220AB only) from terminal to heatsink t = 1 min	V_{AC}	1500				V
Operating junction and storage temperature range	T_J, T_{STG}	-55 to +150				°C

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ELECTRICAL CHARACTERISTICS (T_A= 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP**	**MAX**	**UNIT**
Breakdown voltage	I_R = 1.0 mA	T_A = 25 °C	V_{BR}	150 (min.)	-	V
Instantaneous forward voltage (1)	I_F = 5 A	T_A = 25 °C	V_F	0.69	-	V
	I_F = 10 A			0.84	-	
	I_F = 20 A			1.15	1.43	
	I_F = 5 A	T_A = 125 °C		0.55	-	
	I_F = 10 A			0.64	-	
	I_F = 20 A			0.75	0.82	
Reverse current (2)	V_R = 100 V	T_A = 25 °C	I_R	2	-	µA
		T_A = 125 °C		2.5	-	mA
	V_R = 150 V	T_A = 25 °C		-	250	µA
		T_A = 125 °C		5	25	mA

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 40 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**SYMBOL**	**V20150S**	**VF20150S**	**VB20150S**	**VI20150S**	**UNIT**
Typical thermal resistance	$R_{\theta JC}$	2.0	4.0	2.0	2.0	°C/W

ORDERING INFORMATION (Example)					
PACKAGE	**PREFERRED P/N**	**UNIT WEIGHT (g)**	**PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
TO-220AB	V20150S-E3/4W	1.88	4W	50/tube	Tube
ITO-220AB	VF20150S-E3/4W	1.75	4W	50/tube	Tube
TO-263AB	VB20150S-E3/4W	1.39	4W	50/tube	Tube
TO-263AB	VB20150S-E3/8W	1.39	8W	800/reel	Tape and reel
TO-262AA	VI20150S-E3/4W	1.45	4W	50/tube	Tube

RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

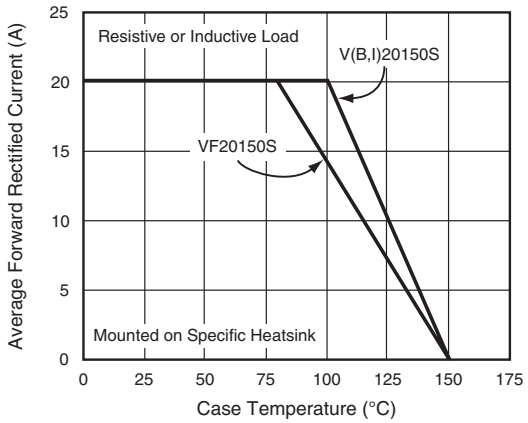


Fig. 1 - Maximum Forward Current Derating Curve

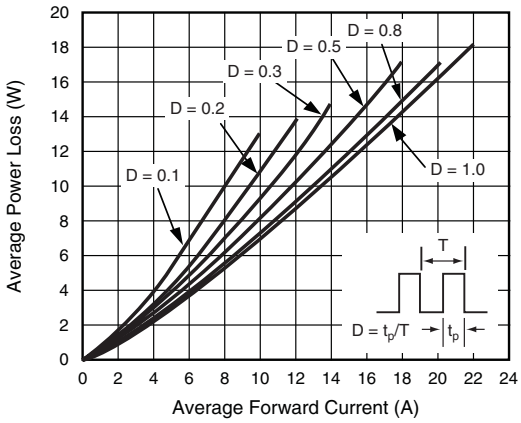


Fig. 2 - Forward Power Loss Characteristics

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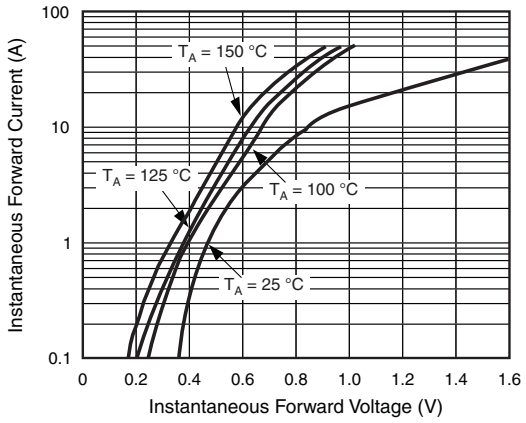


Fig. 3 - Typical Instantaneous Forward Characteristics

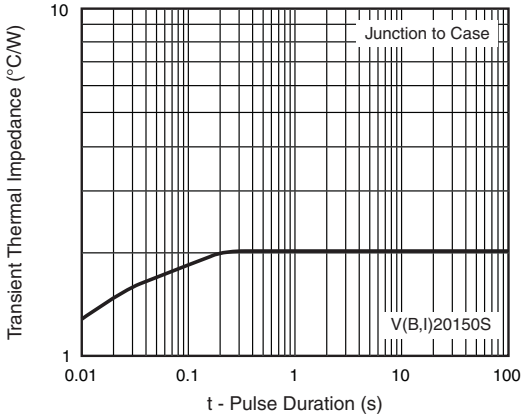


Fig. 6 - Typical Transient Thermal Impedance

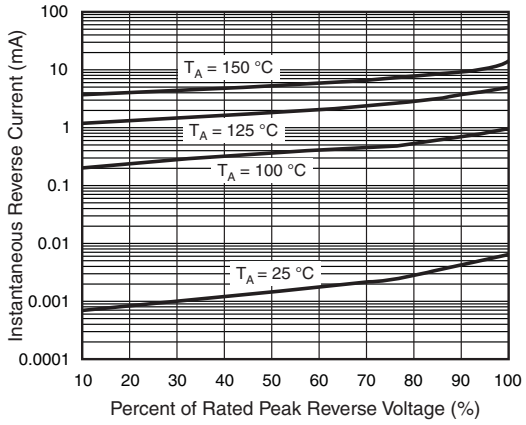


Fig. 4 - Typical Reverse Characteristics

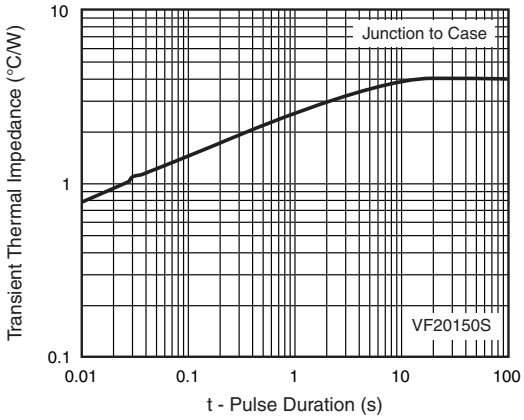


Fig. 7 - Typical Transient Thermal Impedance

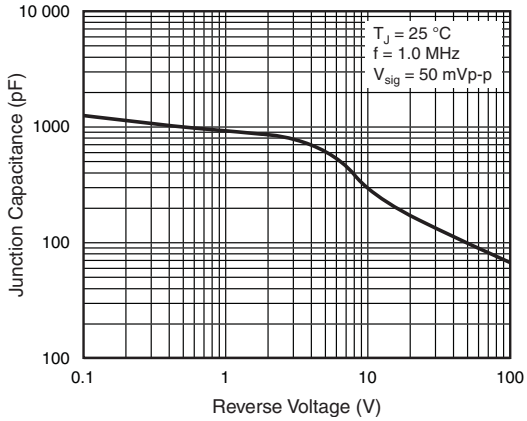
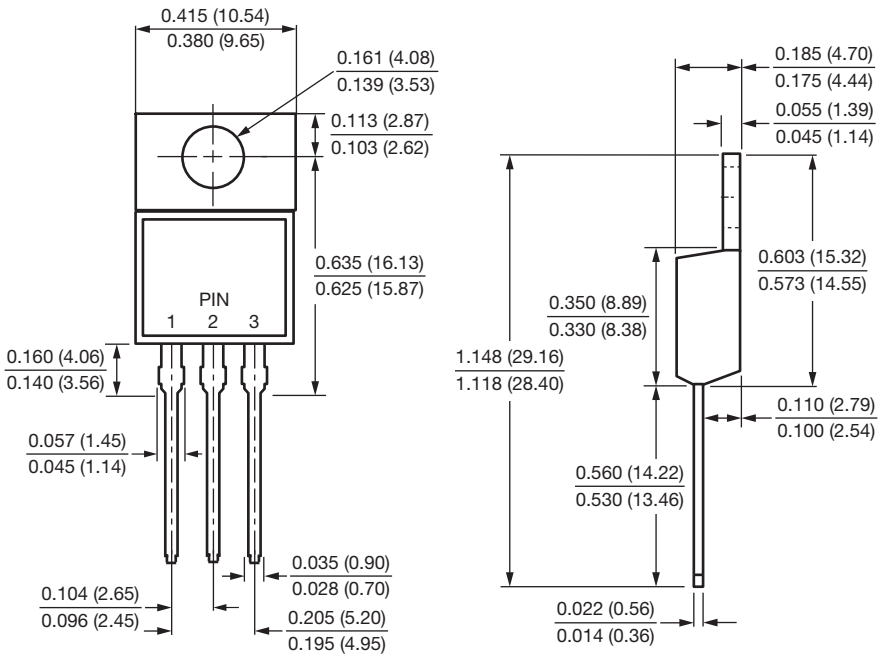


Fig. 5 - Typical Junction Capacitance

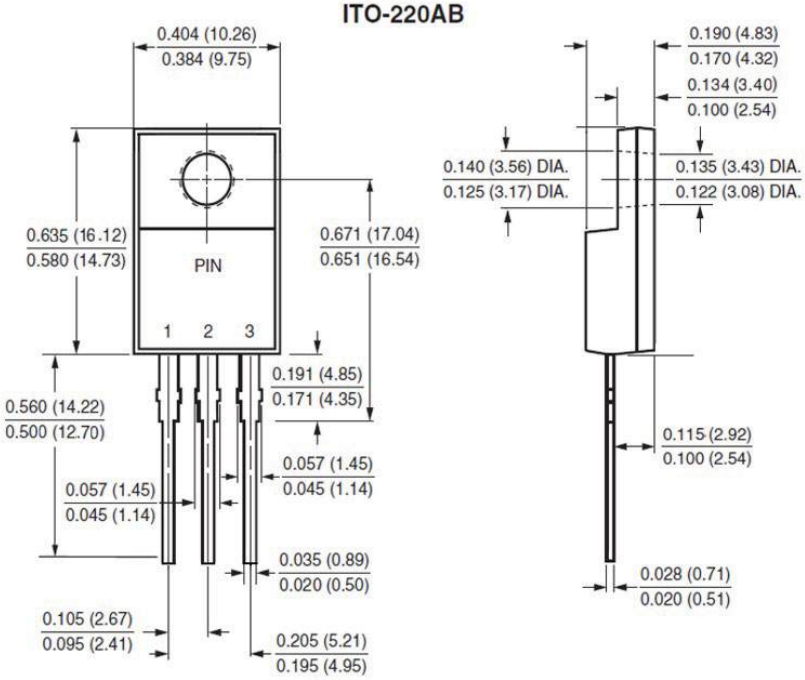
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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

TO-220AB

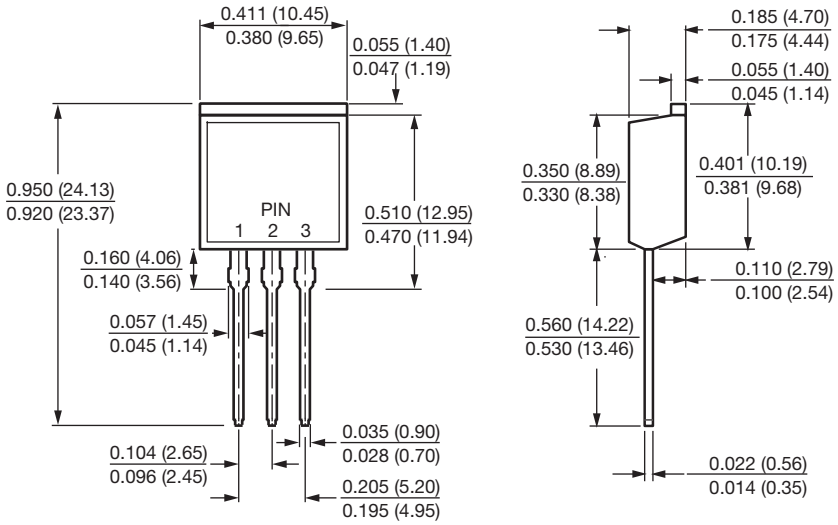


ITO-220AB

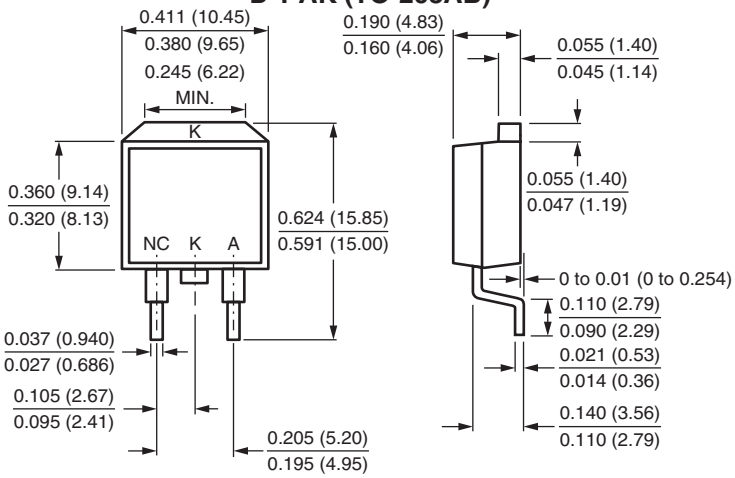


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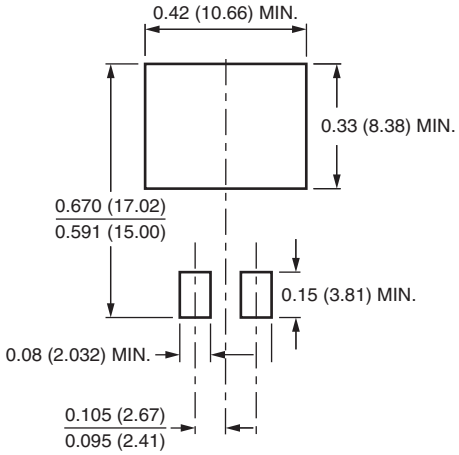
TO-262AA



D²PAK (TO-263AB)



Mounting Pad Layout



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